INCH POUND

MIL-T-27/291C
8 June 2012
SUPERSEDING
MIL-T-27/291B
13 March 1991

MILITARY SPECIFICATION SHEET

# TRANSFORMER, POWER, STEP-DOWN

Inactive for new design after 13 March 1991

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the product described herein shall consist of this document and the latest issue of specification MIL-PRF-27.

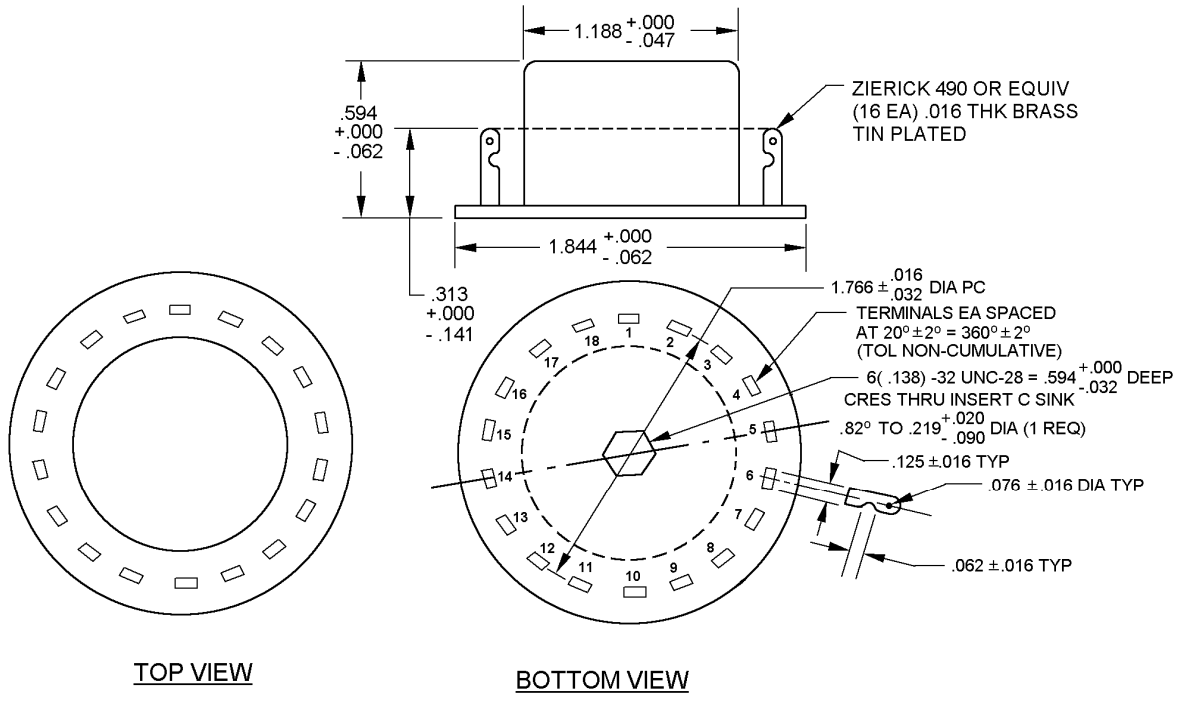


FIGURE 1. Dimensions and configurations.



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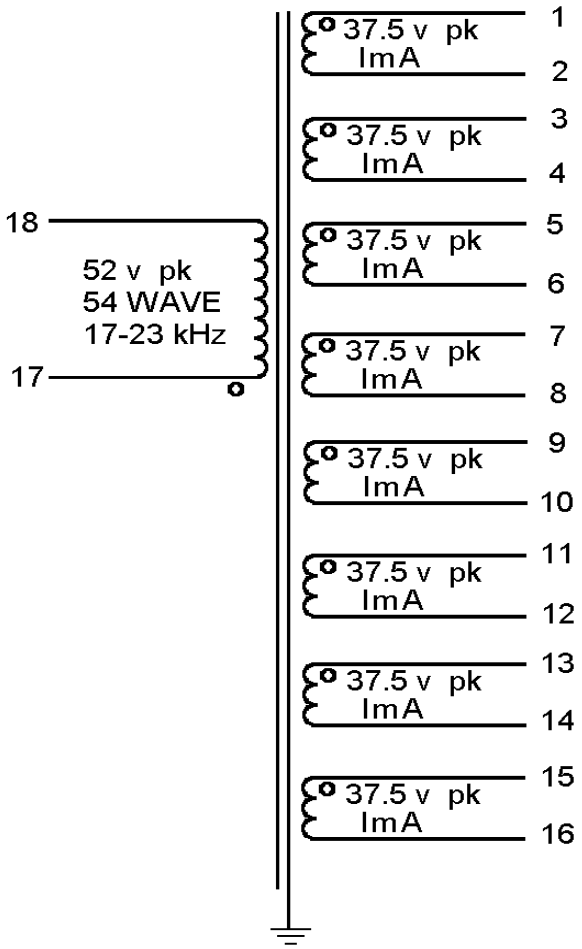


| INCHES | mm |
|---|---|
| .000 | 0.00 |
| .016 | 0.41 |
| .020 | 0.51 |
| .032 | 0.81 |
| .062 | 1.57 |
| .076 | 1.93 |
| .090 | 2.29 |
| .125 | 3.18 |
| .138 | 3.51 |
| .141 | 3.59 |
| .219 | 5.56 |
| .313 | 7.95 |
| .534 | 13.56 |
| .594 | 15.09 |
| 1.188 | 30.18 |
| 1.766 | 44.86 |
| 1.844 | 46.84 |

ALTITUDE: 10,000 FT. MAX
WORKING VOLTAGE:
PRI: (17-18_ 52 V pk
SEC: 37.5 V pk

CIRCUIT DIAGRAM AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Marking shall be on the top of the case.

FIGURE 1. Dimensions and configurations - Continued.



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REQUIREMENTS: (When numbers in parentheses, i.e., (H1-H2) are used, they indicate the winding and the extreme terminals of the windings.)

Electrical ratings:

Primary voltage (11-18): 52 volts pk, 17-23 kHz square wave.

Primary current (17-18): 6 mA, maximum.

Secondary voltage (1-2), (3-4), (5-6), (7-8), (9-10), (11-12), (13-14), (15-16):
37.5 volts peak each winding

Secondary current (1-2), (3-4), (5-6), (7-8), (9-10), (11-12), (13-14), (15-16):
1 mA rms each winding

Working voltage:
(17-18): 52 volts peak.
(1-2), (3-4), (5-6), (7-8), (9-10), (11-12), (13-14), (15-16): 37.5 volts peak.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Material: Epoxy.

Terminals: Brass, solder type lugs.

Terminal height: 0.313 +.000, -.141 inch

Weight: 56.7 grams, maximum.

Operating temperature range: -40°C to +130°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2.0 pounds.

Dielectric withstanding voltage (at sea level):

300 V rms (17-18) to all secondaries and ground.
100 V rms, between secondaries and ground.

Electrical characteristics:

No load: With 52 volts peak, 17-23 kHz across (17-18), the voltage across (1-2), (3-4), (5-6), (7-8), (9-10), (11-12), (13-14), (15-16) shall be 37.5 volts peak ±2 percent.
milliampere and the power in (H1-H2) shall not exceed 2.5 watts.

Rated load: With 52 volts peak, 17-23 kHz across (17-18), the voltage across (1-2), (3-4), (5-6), (7-8), (9-10), (11-12), (13-14), (15-16) shall be 37.5 volts peak ±2 percent.
milliampere and the power in (H1-H2) shall not exceed 2.5 watts.



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DC resistance: (17-18): .67 ohm maximum. (1-2), (3-4), (5-6), (7-8), (9-10), (11-12), (13-14), (15-16): 2.81 ohms maximum each.

Polarity: Additive, with terminals 18 and 1, 2 and 3, 4 and 5, 6 and 7, 8 and 9; 10 and 11, 12 and 13, and 14 and 15, connected.

Temperature rise: 50°C with 52 volts peak, 17 kHz across (17-18) at an ambient temperature of 65°C.

Vibration, high frequency: MIL-STD-202, method 204, test condition A.

Shock (specified pulse): MIL-STD-202, method 213, test condition H.

VERIFICATION:

Qualification inspection: Not applicable for this specification sheet.

Conformance inspection: Groups A and B tests of MIL-PRF-27 shall be applicable.

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/291-01.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202

Custodians:
Army - CR
Navy - EC
Air Force – 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5950-2009-089)

Review activities:
Army – AR, MI
Navy – AS, MC, OS, SH
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at https://assist.daps.dla.mil .



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